

Norway

SLEIPNER

Volve F

F-15

F-15

Design: F-15

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15	Survey Calculation Method:	Minimum Curvature
Design:	F-15	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-15					
Well Position	+N/-S	-3.17 m	Northing:	6,478,560.36 m	Latitude:	58° 26' 29.7063 N
	+E/-W	3.53 m	Easting:	435,053.55 m	Longitude:	1° 53' 15.1494 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-15				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2009	31.12.2008	-2.07	71.61	50,390

Design	F-15				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-3.17	3.53	207.75	

Survey Program	Date	19.07.2010			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
153.40	2,286.70	12 1/4" F-15 Drop Gyro (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD	
2,329.90	3,925.90	8 1/2" F-15 Drop Gyro Revised (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD	
3,963.80	4,042.88	F15 8 1/2" MWD Surveys (F-15)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-3.17	3.53	0.00	0.000	0.000	0.000	
153.40	0.15	253.69	153.40	-3.17	3.52	0.01	0.600	0.600	0.000	
234.20	0.28	146.02	234.20	-3.37	3.53	0.17	0.132	0.048	-39.976	
274.60	0.78	136.21	274.60	-3.65	3.78	0.31	0.376	0.371	-7.285	
315.00	1.46	128.29	314.99	-4.16	4.37	0.49	0.517	0.505	-5.881	
355.30	3.41	124.07	355.25	-5.15	5.77	0.71	1.457	1.452	-3.141	
395.70	5.13	126.98	395.54	-6.91	8.21	1.14	1.287	1.277	2.161	
436.10	5.52	139.07	435.76	-9.47	10.92	2.13	0.880	0.290	8.978	
474.40	5.62	152.33	473.89	-12.52	13.00	3.87	1.009	0.078	10.386	

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Design:	F-15	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
514.80	6.19	163.05	514.07	-16.36	14.55	6.54	0.920	0.423	7.960
557.10	7.03	171.79	556.09	-21.10	15.59	10.26	0.928	0.596	6.199
597.50	7.74	174.05	596.16	-26.25	16.22	14.52	0.570	0.527	1.678
637.80	8.49	174.54	636.05	-31.91	16.79	19.27	0.561	0.558	0.365
678.20	8.94	175.57	675.99	-38.01	17.31	24.42	0.354	0.334	0.765
718.50	9.34	177.45	715.77	-44.40	17.70	29.89	0.372	0.298	1.400
758.90	10.22	185.51	755.59	-51.24	17.50	36.04	1.207	0.653	5.985
799.30	11.89	193.59	795.24	-58.86	16.18	43.40	1.689	1.240	6.000
839.70	13.68	199.13	834.64	-67.42	13.64	52.16	1.610	1.329	4.114
880.00	15.14	201.54	873.67	-76.81	10.14	62.10	1.175	1.087	1.794
920.40	13.72	203.30	912.79	-86.12	6.31	72.12	1.103	-1.054	1.307
960.80	14.53	201.31	951.97	-95.24	2.57	81.94	0.701	0.601	-1.478
1,001.10	16.51	200.62	990.80	-105.31	-1.28	92.64	1.480	1.474	-0.514
1,041.60	16.33	201.19	1,029.65	-116.01	-5.37	104.01	0.179	-0.133	0.422
1,081.90	16.18	205.25	1,068.34	-126.37	-9.81	115.25	0.853	-0.112	3.022
1,122.30	17.54	211.88	1,107.01	-136.63	-15.43	126.94	1.747	1.010	4.923
1,162.60	17.66	221.21	1,145.43	-146.39	-22.66	138.95	2.100	0.089	6.945
1,203.00	16.05	230.28	1,184.10	-154.57	-31.00	150.07	2.286	-1.196	6.735
1,243.30	15.33	232.77	1,222.90	-161.35	-39.52	160.04	0.734	-0.536	1.854
1,283.80	18.17	237.03	1,261.68	-168.03	-49.09	170.40	2.291	2.104	3.156
1,324.10	21.21	241.08	1,299.62	-174.98	-60.74	181.98	2.479	2.263	3.015
1,364.50	24.71	243.03	1,336.81	-182.34	-74.67	194.98	2.659	2.599	1.448
1,404.90	24.39	243.20	1,373.56	-189.94	-89.64	208.67	0.243	-0.238	0.126
1,445.20	26.03	238.06	1,410.02	-198.37	-104.57	223.08	2.035	1.221	-3.826
1,485.60	28.35	233.84	1,445.96	-208.72	-119.85	239.35	2.239	1.723	-3.134
1,525.90	28.46	236.92	1,481.41	-219.61	-135.62	256.33	1.094	0.082	2.293
1,566.30	28.53	240.98	1,516.92	-229.54	-152.12	272.81	1.439	0.052	3.015
1,606.60	28.60	242.01	1,552.31	-238.74	-169.05	288.83	0.370	0.052	0.767
1,647.00	28.52	239.48	1,587.80	-248.17	-185.90	305.03	0.900	-0.059	-1.879
1,687.30	28.58	236.33	1,623.20	-258.40	-202.21	321.67	1.121	0.045	-2.345
1,727.70	28.59	233.29	1,658.68	-269.54	-218.01	338.88	1.080	0.007	-2.257
1,768.10	28.37	229.11	1,694.19	-281.60	-233.01	356.54	1.489	-0.163	-3.104
1,808.40	28.17	223.89	1,729.69	-294.73	-246.85	374.60	1.846	-0.149	-3.886
1,848.80	28.30	224.77	1,765.28	-308.40	-260.20	392.92	0.324	0.097	0.653
1,889.10	28.48	221.22	1,800.74	-322.41	-273.26	411.40	1.263	0.134	-2.643
1,939.50	28.59	217.77	1,845.02	-340.98	-288.57	434.96	0.983	0.065	-2.054
1,969.90	28.59	214.37	1,871.72	-352.74	-297.13	449.35	1.605	0.000	-3.355
2,010.20	29.15	214.06	1,907.01	-368.83	-308.07	468.68	0.432	0.417	-0.231
2,050.60	29.23	214.38	1,942.28	-385.12	-319.15	488.26	0.130	0.059	0.238
2,090.90	30.12	215.13	1,977.29	-401.51	-330.53	508.06	0.718	0.663	0.558
2,131.30	30.25	213.42	2,012.21	-418.29	-341.97	528.24	0.646	0.097	-1.270
2,171.60	30.35	210.52	2,047.01	-435.54	-352.73	548.51	1.092	0.074	-2.159
2,212.00	32.35	207.87	2,081.51	-453.89	-362.97	569.52	1.804	1.485	-1.968

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Design:	F-15	Database:	Production EDM P246N

Survey										
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2,252.30	34.65	207.17	2,115.12	-473.62	-373.24	591.76	1.736	1.712	-0.521	
2,286.70	34.23	206.97	2,143.49	-490.94	-382.09	611.22	0.379	-0.366	-0.174	
2,329.90	32.44	206.19	2,179.58	-512.17	-392.72	634.95	1.278	-1.243	-0.542	
2,370.20	33.22	205.79	2,213.44	-531.81	-402.29	656.79	0.603	0.581	-0.298	
2,410.60	33.23	205.61	2,247.24	-551.75	-411.89	678.91	0.074	0.007	-0.134	
2,450.90	33.19	205.41	2,280.95	-571.67	-421.40	700.96	0.087	-0.030	-0.149	
2,491.30	33.08	205.67	2,314.78	-591.60	-430.92	723.03	0.133	-0.082	0.193	
2,531.70	32.85	207.37	2,348.68	-611.27	-440.73	745.01	0.708	-0.171	1.262	
2,572.10	34.69	206.81	2,382.26	-631.26	-450.95	767.46	1.386	1.366	-0.416	
2,612.40	38.13	206.72	2,414.69	-652.61	-461.72	791.37	2.561	2.561	-0.067	
2,652.80	42.06	206.90	2,445.59	-675.83	-473.46	817.39	2.920	2.918	0.134	
2,693.20	46.09	206.27	2,474.61	-700.96	-486.03	845.48	3.010	2.993	-0.468	
2,733.60	50.71	203.11	2,501.43	-728.41	-498.61	875.63	3.853	3.431	-2.347	
2,773.90	55.83	201.10	2,525.52	-758.33	-510.75	907.75	3.995	3.811	-1.496	
2,814.30	59.21	200.84	2,547.21	-790.15	-522.94	941.59	2.515	2.510	-0.193	
2,854.60	61.44	201.50	2,567.16	-822.80	-535.59	976.37	1.714	1.660	0.491	
2,895.00	63.80	200.63	2,585.74	-856.27	-548.48	1,012.00	1.844	1.752	-0.646	
2,935.40	64.92	201.01	2,603.22	-890.31	-561.42	1,048.16	0.870	0.832	0.282	
2,975.80	65.81	201.56	2,620.06	-924.53	-574.75	1,084.65	0.758	0.661	0.408	
3,016.20	67.69	201.29	2,636.01	-959.08	-588.31	1,121.54	1.408	1.396	-0.200	
3,056.50	67.51	202.60	2,651.37	-993.64	-602.23	1,158.60	0.912	-0.134	0.975	
3,095.90	67.31	202.55	2,666.50	-1,027.23	-616.20	1,194.83	0.156	-0.152	-0.038	
3,137.20	65.72	201.61	2,682.96	-1,062.33	-630.44	1,232.53	1.314	-1.155	-0.683	
3,177.60	65.37	202.26	2,699.69	-1,096.44	-644.18	1,269.11	0.510	-0.260	0.483	
3,217.90	66.49	203.76	2,716.12	-1,130.31	-658.56	1,305.78	1.317	0.834	1.117	
3,258.30	65.99	203.55	2,732.40	-1,164.18	-673.40	1,342.66	0.398	-0.371	-0.156	
3,298.60	66.24	204.25	2,748.72	-1,197.87	-688.33	1,379.43	0.512	0.186	0.521	
3,339.00	66.42	205.60	2,764.94	-1,231.42	-703.92	1,416.38	0.928	0.134	1.002	
3,379.40	66.10	204.95	2,781.20	-1,264.86	-719.71	1,453.33	0.502	-0.238	-0.483	
3,419.70	66.87	201.89	2,797.29	-1,298.77	-734.39	1,490.17	2.166	0.573	-2.278	
3,460.10	65.46	201.03	2,813.61	-1,333.16	-747.91	1,526.90	1.199	-1.047	-0.639	
3,500.50	65.16	201.35	2,830.49	-1,367.38	-761.18	1,563.37	0.310	-0.223	0.238	
3,540.80	65.48	203.11	2,847.32	-1,401.28	-775.03	1,599.82	1.214	0.238	1.310	
3,581.20	65.70	202.95	2,864.01	-1,435.13	-789.43	1,636.48	0.196	0.163	-0.119	
3,621.50	66.91	199.87	2,880.21	-1,469.48	-802.89	1,673.15	2.284	0.901	-2.293	
3,661.90	62.82	197.67	2,897.37	-1,504.10	-814.67	1,709.27	3.378	-3.037	-1.634	
3,702.30	57.41	199.75	2,917.49	-1,537.27	-825.88	1,743.84	4.234	-4.017	1.545	
3,742.60	55.25	200.14	2,939.83	-1,568.79	-837.32	1,777.07	1.626	-1.608	0.290	
3,783.00	53.10	199.72	2,963.48	-1,599.58	-848.49	1,809.52	1.616	-1.597	-0.312	
3,823.30	51.77	201.86	2,988.05	-1,629.45	-859.82	1,841.22	1.605	-0.990	1.593	
3,863.60	51.32	205.40	3,013.12	-1,658.35	-872.46	1,872.69	2.091	-0.335	2.635	
3,904.00	48.01	210.10	3,039.27	-1,685.60	-886.76	1,903.47	3.621	-2.458	3.490	
3,925.90	47.25	210.48	3,054.03	-1,699.57	-894.92	1,919.63	1.110	-1.041	0.521	

Statoil

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3,963.80	45.54	207.99	3,080.17	-1,723.51	-908.33	1,947.06	1.967	-1.354	-1.971
4,002.89	44.85	204.69	3,107.72	-1,748.36	-920.64	1,974.78	1.873	-0.530	-2.533
4,042.88	44.32	202.03	3,136.20	-1,774.12	-931.77	2,002.76	1.456	-0.398	-1.995
4,090.00	44.32	202.03	3,169.92	-1,804.64	-944.12	2,035.52	0.000	0.000	0.000

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.60	220.60	30"	30.000	36.000	
1,368.40	1,340.35	20"	20.000	26.000	
2,534.00	2,350.61	9 5/8"	9.625	12.250	

Checked By: _____ Approved By: _____ Date: _____